

Distributed Inference Network(DIN) & Internet of Agents (IOA)

Side Meeting

IETF 123 Madrid

Note Well

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- BCP 25 (Anti-Harassment Procedures)
 - BCP 54 (Code of Conduct)
 - BCP 78 (Copyright)
- BCP 79 (Patents, Participation)
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Motivation

- The explosive growth in AI inference services marks the inclusive AI inference era, presenting both challenges and opportunities for networking infrastructure.
- Centralized deployment inference architectures struggle to scale for massive concurrency (users/API/M2M/A2A) + stringent low latency demands.
- This urgently requires novel Distributed Inference Network (DIN) architectures to handle extreme concurrency and latency, and research in scenarios, requirements and technologies.
- As AI evolves: *Human-model* → *Multi Agent System* → *Internet of Agents (IOA)*, DIN emerges as the critical infrastructure layer.
- These necessitates proactive explorations and development by networking professionals to meet the unique demands of evolving landscape.
- This meeting is about promoting broader thinking on how distributed network can accelerate the AI inference and IOA.

Agenda

	Topic	Time(mins)	Presenter	Company
0	Administration & Review	5	Shengnan Yue	China Mobile
1	Distributed Inference Network (DIN) - Scenarios, Requirements, and Experiments	15	Shengnan Yue	China Mobile
2	Internet of Agent and DIN - Opportunities and Challenges	15	Shuping Peng	Huawei
3	The Model Scheduling of Distributed Inference Network (DIN)	10	Haiyang Zhang	H3C
4	Open Discussion	30	All	

Questions List

- Questions to Have in Mind You don't have to answer these questions, but please think about them for the discussion at the end of the meeting. (not limited to these questions)
 1. What services can carrier networks provide to accelerate inference in the era of inclusive AI inference, and what scenarios and new technologies should be developed?
 2. What inference network technologies will accommodate AI evolution into Internet of Agents (IOA) ?
 3. What new protocols are needed, or what are the opportunities for protocol optimization, in the era of distributed inference and IOA?
 4. Further actions?